

Gujarat University
Department of Computer Science
Master of Computer Applications (MCA) – Semester 3
Subject: Computer Vision

Practical Assignment – 2

Practical Questions

Question 1 – Canny Edge Detection

Write a Python program to:

- Read an input image.
- Convert it to grayscale.
- Apply Gaussian smoothing.
- Apply Canny edge detection.
- Display the original and edge-detected images.

Try different lower and upper threshold values and observe the changes.

Question 2 – Laplacian of Gaussian (LoG)

Write a Python program to:

- Read an image.
- Convert it to grayscale.
- Apply Gaussian Blur.
- Apply the Laplacian operator.
- Display the detected edges.

Change the Gaussian kernel size and observe the effect.

Question 3 – Difference of Gaussian (DoG)

Write a Python program to:

- Apply Gaussian blur using two different sigma values.
- Subtract the two blurred images.
- Display the resulting DoG image.
- Compare the DoG result with the original image.

Question 4 – Comparison of Edge Detectors

Apply Canny, LoG, and DoG to the same image. Display all results and compare:

- Edge quality
- Noise
- Number of detected edges
- Computational requirements

Question 5 – Hough Line Transform

Write a Python program to:

- Detect edges using Canny.
- Apply the Standard Hough Line Transform.

- Detect straight lines.
- Draw the detected lines on the original image.

Question 6 – Probabilistic Hough Transform

Apply `HoughLinesP()` to an edge-detected image and draw the detected line segments.

Change the threshold, minimum line length, and maximum line gap.

Question 7 – Hough Transform for Road/Lane Detection

Use a road image and:

- Detect edges using Canny.
- Apply the Probabilistic Hough Transform.
- Detect lane lines.
- Draw the detected lines on the road image.

Question 8 – Harris Corner Detection

Write a Python program to:

- Convert an image to grayscale.
- Apply Harris Corner Detection.
- Apply a suitable threshold.
- Mark the detected corners on the image.

Question 9 – Harris Corner Parameter Analysis

Apply Harris Corner Detection using different values of:

- Block size
- Sobel aperture size
- Harris detector parameter k
- Compare the detected corners.

Question 10 – Harris Corner Detection on Shapes

Create an image containing squares, rectangles, and triangles. Apply Harris Corner Detection and mark the detected corners.

Question 11 – Hessian-Based Interest Point Detection

Write a program to calculate the Hessian matrix/determinant at different image locations and identify strong interest points. Display the detected points on the image.

Question 12 – Hessian-Affine Feature Detection

Detect interest regions using a Hessian-Affine based method or a suitable computer-vision library. Display the detected keypoints/regions and explain how the method provides robustness to scale and affine changes.

Question 13 – Image Gradient and Orientation

Use Sobel operators to calculate:

- G_x and G_y
- Gradient magnitude
- Gradient orientation
- Display the resulting images.

Question 14 – Orientation Histogram

Divide an image into small cells and:

- Calculate gradient magnitude and orientation.
- Divide orientations into bins.
- Construct an orientation histogram for each cell.
- Display the histogram.

Question 15 – SIFT Keypoint Detection

Write a Python program to:

- Read an image.
- Create a SIFT detector.
- Detect keypoints.
- Extract descriptors.
- Draw the keypoints on the image.
- Display the number of detected keypoints.

Question 16 – SIFT Feature Matching

Take two images containing the same object. Use SIFT to:

- Detect keypoints.
- Extract descriptors.
- Match descriptors using BFMatcher.
- Display the matched keypoints.

Question 17 – SIFT and Scale Variation

Resize the same image to different scales and apply SIFT to each image. Compare the detected keypoints and explain why SIFT is useful for scale changes.

Question 18 – SURF Feature Detection

Use a suitable OpenCV/contributed implementation to:

- Detect SURF keypoints.
- Extract descriptors.
- Draw the keypoints.
- Report the number of detected keypoints.

Question 19 – SIFT vs SURF

Apply SIFT and SURF to the same image and compare:

- Number of keypoints
- Descriptor size
- Processing time
- Scale robustness
- Rotation robustness
- Present your observations in a table.

Question 20 – HOG Feature Extraction

Write a Python program to:

- Resize the image.
- Calculate image gradients.
- Divide the image into cells.
- Calculate orientation histograms.
- Create HOG features.
- Display the HOG visualization.

Question 21 – HOG for Human Detection

Use the OpenCV HOG descriptor and its pre-trained people detector to detect humans in an image. Draw bounding boxes around detected persons.

Question 22 – GLOH Feature Descriptor

Implement or use an appropriate library implementation of GLOH to extract local image features. Explain:

- Spatial regions
- Orientation bins
- Feature vector formation
- Difference between GLOH and SIFT

Question 23 – SIFT vs GLOH

Extract SIFT and GLOH features from the same image and compare:

- Descriptor representation
- Feature dimension
- Computational cost
- Robustness
- Applications

Question 24 – Complete Feature Detection Pipeline

For a given image, apply the following pipeline:

- Image → Grayscale → Gaussian Blur → Canny → Harris → SIFT → HOG
- Display the output of each stage and briefly explain the purpose of each technique.

Question 25 – Comparative Study of Computer Vision Features

For the same image, apply the following techniques and prepare a comparison based on type, what is detected/described, scale invariance, rotation invariance, and main application:

- Canny
- LoG
- DoG
- Hough Transform
- Harris
- Hessian-Affine
- SIFT
- SURF
- HOG
- GLOH

Technique	Type	Detects/Describes	Scale Invariant	Rotation Invariant	Main Application
Canny	Edge detector	Edges	No	No	Edge detection
LoG	Edge detector	Edges	Limited	No	Edge detection
DoG	Feature detector	Blobs/features	Yes	No	Feature detection
Hough	Line detector	Lines	No	No	Line detection
Harris	Corner detector	Corners	No	Partially	Corner detection
Hessian-Affine	Interest point	Regions/features	Yes	Yes	Feature detection
SIFT	Descriptor	Keypoints/features	Yes	Yes	Matching/recognition
SURF	Descriptor	Keypoints/features	Yes	Yes	Matching/recognition
HOG	Descriptor	Shape/gradient	Limited	Limited	Object/person detection
GLOH	Descriptor	Local features	Yes	Yes	Feature matching